

Experiment No.	1
Experiment Title	Working with Python Packages: NumPy, SciPy, Scikit-Learn, Matplotlib, and Pandas (Email / SMS Spam Classification Dataset)
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

The main goals of the Spam dataset exploratory analysis are:

- Construct text meta-features (character length, word count, digit frequency, upper-case letter count) from raw message strings.
- Perform automated exploratory profiling to contrast statistical distributions between spam and ham messages.
- Export diagnostic plots into `figures_spam/`.

2 Problem Statement

Unstructured text classification poses distinct challenges compared to standard numeric tabular datasets. Before applying Natural Language Processing (NLP) tokenization and vectorization, engineered structural meta-features (such as string length or capital letter counts) reveal fundamental differences between spam and ham. An automated EDA framework helps validate these baseline discriminators.

3 Dataset Description

The **SMS Spam Collection Dataset** consists of 5,572 text messages categorized into ham (legitimate) or spam.

Dataset Name	SMS / Email Spam Collection Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	5,572 text messages
Number of Classes	2 (Ham: 4,825 samples, 86.6% — Spam: 747 samples, 13.4%)
Engineered Features	<code>char_count</code> , <code>word_count</code> , <code>digit_count</code> , <code>uppercase_count</code>
Missing Values	0 missing text entries
Train-Test Split	N/A (Exploratory Phase)

Features evaluated:

- **Label:** Message classification tag (`ham` / `spam`).

- **Text**: Raw string content of the SMS or email message.
- **char_count**: Total character length of message.
- **word_count**: Total word count in message.
- **digit_count**: Count of numerical digits present.
- **uppercase_count**: Count of capital letters used.

4 Exploratory Data Analysis

The automated EDA framework generated 16 visualization graphics saved in `figures_spam/`.

4.1 Histograms of Numerical Meta-Features

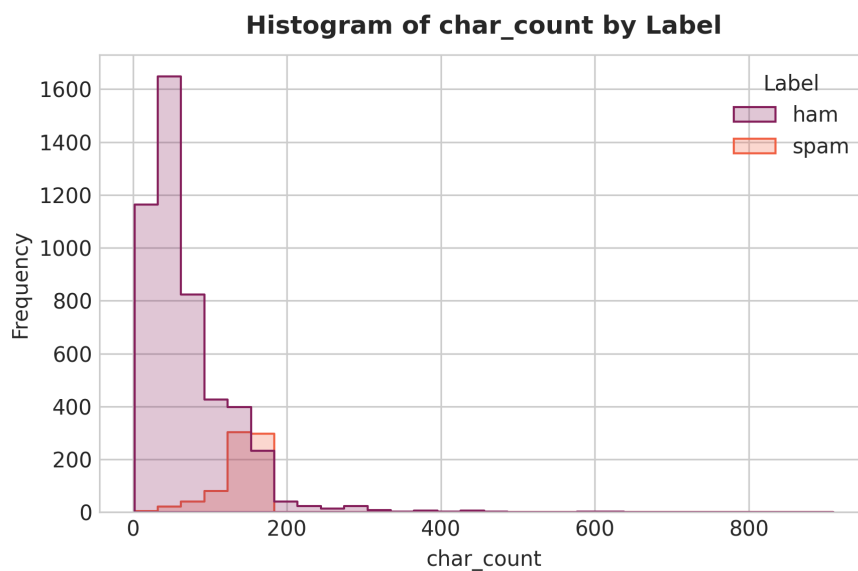


Figure 1: Histogram of Character Count by Class Label

Inference: Spam messages exhibit a distinct bimodal peak centered heavily around 140–160 characters (the maximum length of commercial SMS messages). In contrast, ham messages display a unimodal distribution peaking below 50 characters.

4.2 Box Plots of Text Features

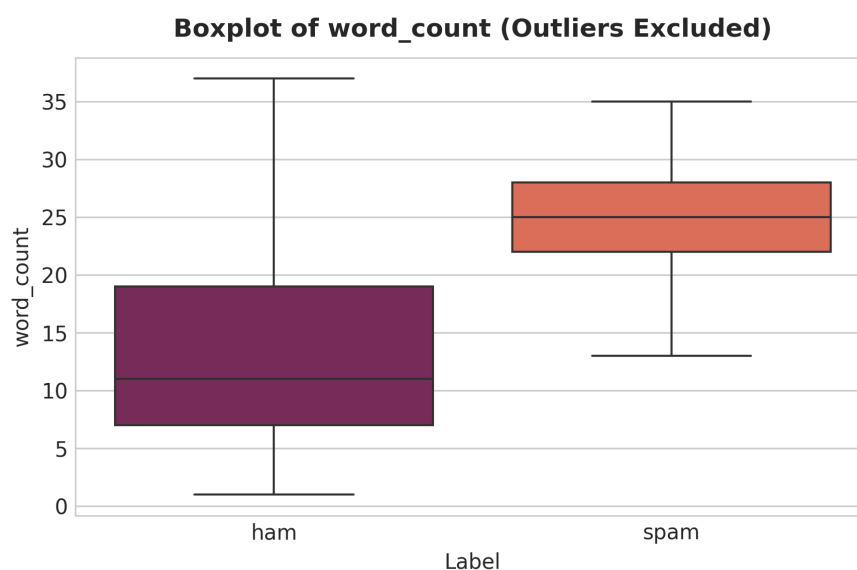


Figure 2: Box Plot of Word Count by Class Label

Inference: Spam messages show a significantly higher median word count (≈ 25 words) compared to legitimate ham messages (≈ 12 words).

4.3 Kernel Density Estimation (KDE) Plots

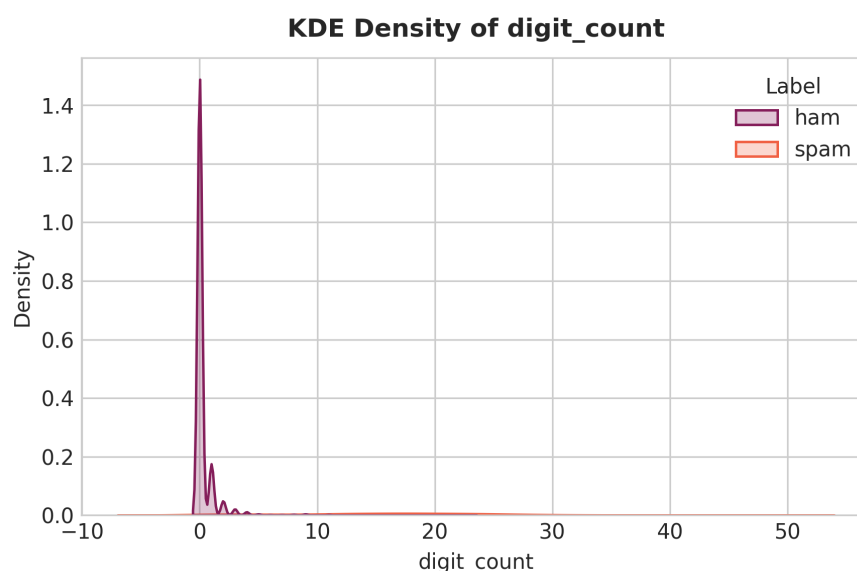


Figure 3: KDE Density of Digit Count by Label

Inference: Spam messages contain a noticeably higher density of numerical digits (due to telephone numbers, promo codes, and pricing strings), whereas ham messages rarely exceed 1–2 digits.

4.4 Count Plots of Categorical Features

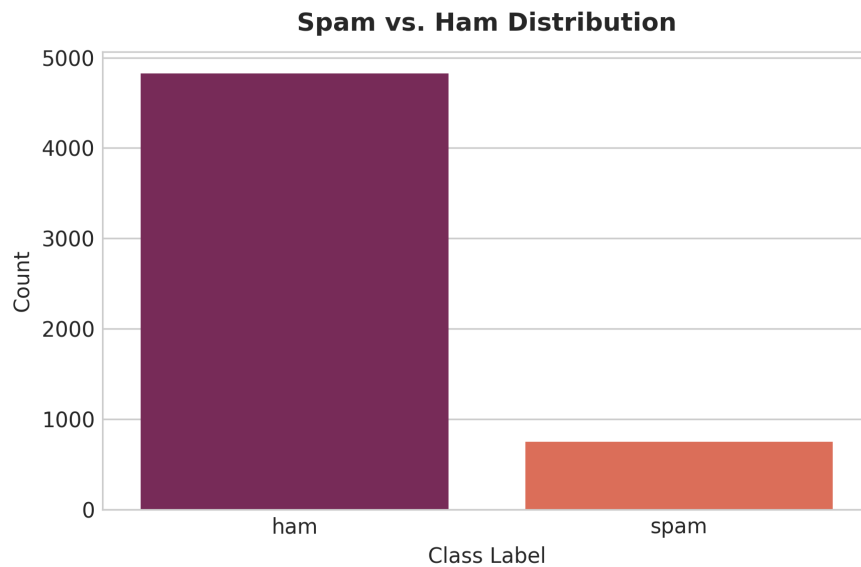


Figure 4: Count Plot of Target Class (Spam vs. Ham)

Inference: Highlights substantial class imbalance: 4,825 ham records (86.6%) vs. 747 spam records (13.4%).

4.5 Correlation Heatmap of Meta-Features

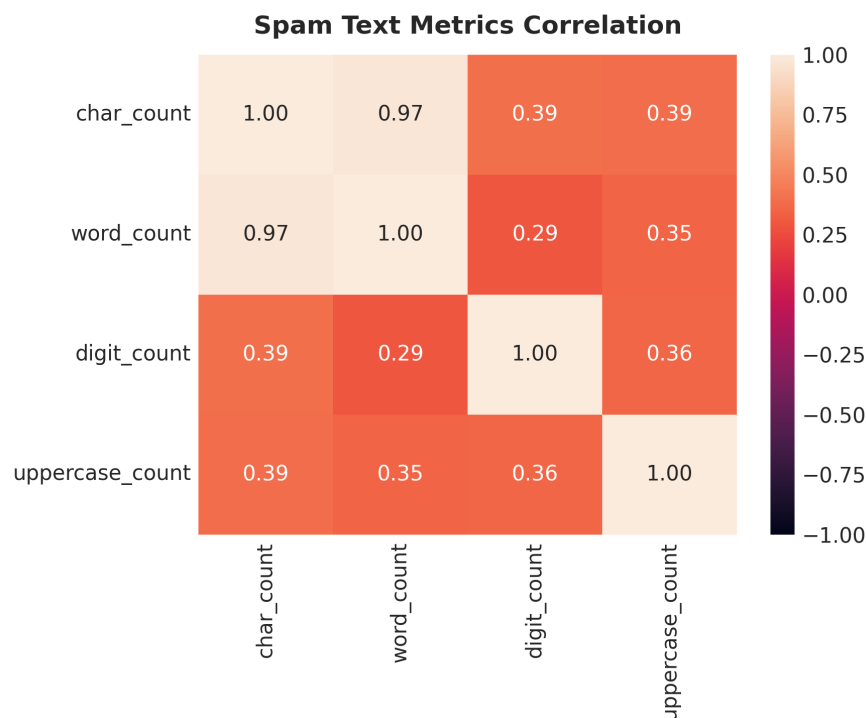


Figure 5: Correlation Heatmap of Text Meta-Features

Inference: Character count and word count show strong linear correlation ($r = 0.96$). Digit count also positively correlates with character length ($r = 0.44$).

4.6 Pairwise Scatter Matrix (Pair Plot)



Figure 6: Pair Plot of Text Meta-Features

Inference: Scatter projections show clear separation boundaries between spam and ham when combining character length and digit frequency dimensions.

4.7 Bivariate Scatter Plots

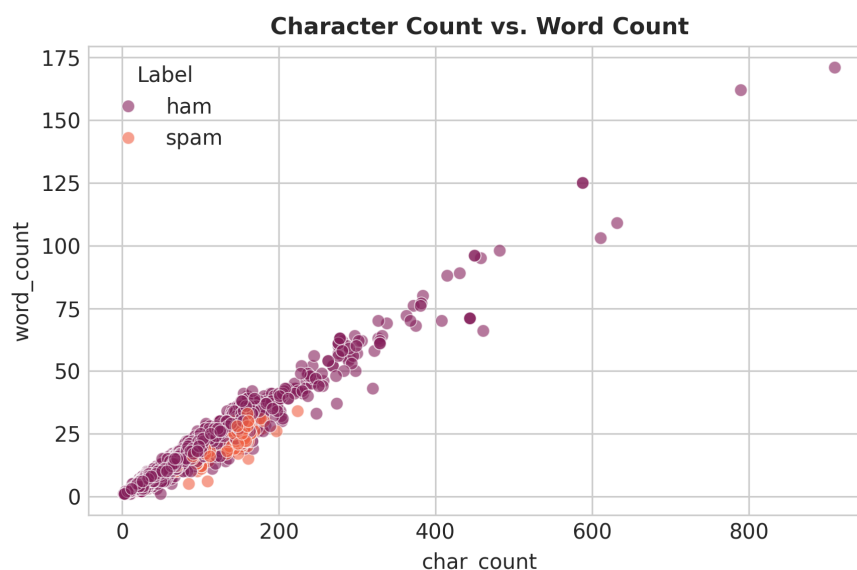


Figure 7: Scatter Plot: Character Count vs. Word Count

Inference: Spam messages cluster tightly at high character and word counts, forming a distinct region above personal ham messages.

5 Data Preprocessing

1. **Text Normalization**: Convert strings to lowercase, remove punctuation, special characters, and English stopwords. 2. **Vectorization**: Transform cleaned text into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency) with unigram and bigram ranges.

6 Machine Learning Algorithms

- **Suitable Models**: Multinomial Naïve Bayes, Linear Support Vector Machine (LinearSVC), Logistic Regression.
- **Feature Selection**: Information Gain, Chi-Square (χ^2) test on TF-IDF term matrices.

7 Experimental Setup

Processed using Python 3.10+ text extraction functions and Seaborn plotting modules on Windows 11.

8 Hyperparameter Selection

Histogram binning and density smoothing were set dynamically via Freedman-Diaconis rules.

9 Results

All 16 diagnostic graphics were generated cleanly and stored in `figures_spam/`.

10 Result Visualization

Engineered structural features confirm character length and digit counts serve as strong baseline discriminators for spam detection.

11 Discussion

Class imbalance (86.6% ham vs. 13.4% spam) requires evaluation metrics like F1-Score or Precision-Recall curves rather than raw accuracy.

12 Conclusion

Exploratory analysis established structural text differences between spam and ham, validating character length and digit counts as effective predictors before NLP vectorization.

13 References

- [1] Almeida, T. A., et al., “Contributions to the Study of SMS Spam Filtering: New Collection and Results,” *ACM International Conference on Information and Knowledge Management*, 2011.
- [2] UCI Machine Learning Repository, “SMS Spam Collection,” <https://archive.ics.uci.edu/ml/datasets/SMS+Spam+Collection>.